

Deep Reinforcement Learning for Adaptive Gain Tuning in Control of Teleoperation Manipulators with Joint Flexibility and Time-Varying Delays

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Abstract—Bilateral teleoperation systems that include joint flexibility better reflect real robotic systems used in surgery, space, and rehabilitation. However, joint flexibility together with time-varying communication delays makes it difficult to maintain stable and coordinated motion between the master and slave robots. To address this, we propose a hybrid control method that combines a stable Proportional-plus-Damping (P+d) controller with a model-free deep reinforcement learning agent based on the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm. The P+d controller provides basic stability under bounded delays, while the learning agent adjusts and tunes the remote-side proportional and damping gains in real time to reduce vibrations and improve tracking. Stability is guaranteed for bounded time-varying delays using Lyapunov-Krasovskii analysis. The approach provides a practical solution for teleoperation systems facing both joint flexibility and uncertain network delays.

Index Terms—Bilateral Teleoperation, Deep Reinforcement Learning, Joint Flexibility, Time-varying Delays, Adaptive Gain Tuning

I. INTRODUCTION

A teleoperation manipulator system consists of a human operator, a local master robot, a communication channel, and a remote slave robot. The remote robot operates within a distant environment. Teleoperation enables human manipulation and sensory capabilities in remote or hazardous locations. The system is designed to ensure stable and transparent interaction between the master and slave robots. Teleoperation and robotic control technologies have facilitated substantial advancements in domains that require remote manipulation, including space exploration [1], rehabilitation [2], and surgery [3]. These application areas underscore the need for advanced teleoperation systems capable of managing complex and high-risk procedures. Commercial platforms such as the da Vinci surgical system [4] and the Sina teleoperation system [5] exemplify practical implementations, highlighting the continuous development of teleoperation technologies. Robotics research originated with Goertz's development of the first master-slave manipulator in the 1940s, which established bilateral teleoperation. In subsequent decades, researchers addressed key challenges in remote robotic interaction, including stability, delay compensation, and transparency. These efforts encompassed

supervisory and software-based control, passivity theory, scattering methods, and Internet-based architectures [6]. These issues have motivated the control theoretic research in teleoperation over the past decades. The scattering transformation, introduced by Anderson and Spong [7] in the late 1980s, has played a key role in teleoperation system control. However, most methods based on this approach experience position drift, with only a few exceptions [8]. To overcome this, Chopra et al. [9] developed adaptive control schemes that reduce position drift without using scattering variables. Similarly, Nuño et al. presented another adaptive strategy that synchronizes the positions of master and slave manipulators even when constant time delays are present, and later provided a corrected stability proof [10]. Recent research has focused on overcoming stability and transparency challenges in bilateral teleoperation systems operating under constant communication delays. For example, Ganjefar et al. [11] proposed the utilization of wave-variables and the Smith predictor as strategies to minimize the impact of delays. Venkateswaran and Qu [12] introduced a passivity-shortage-based control framework. This framework extends conventional passivity theory to systems with constant delays. Unlike the classical scattering or wave-variable methods, their design uses a negative-feedback interconnection of passivity-short systems. This approach guarantees L^2 stability even when the delay violates strict passivity [12]. Lu et al. [13] proposed an adaptive bilateral controller based on the position-error-based (PEB) structure to handle uncertain dynamics and constant communication delay. Through Lyapunov analysis, they proved global stability and verified the controller on a 2-DOF master-slave manipulator in simulation. Furthermore, the adaptive law compensated for parameter uncertainty and achieved accurate position tracking [13]. Wang et al. [14] combined terminal sliding-mode control with neural-network adaptation to obtain finite-time convergence in a force-feedback teleoperation system under constant delay. The Lyapunov-based proof guaranteed boundedness of all signals and ensured that master-slave tracking errors vanished in finite time [14]. Yang et al. [15] developed a wave-variable compensator for a three-channel (3CH) bilateral teleoperation system with constant communication delay. They reformulated the 3CH architecture into a two-port passive network. Then, they introduced two wave-variable compensators with energy

reservoirs. This approach minimized the bias that normally degrades transparency [15]. Uyulan [16] introduced a robust passivity-based nonlinear controller that incorporates high-order sliding-mode observers to address load disturbances and communication delays. This method preserved passivity and transparency by compensating for disturbances using super-twisting observers. In practical teleoperation systems, time-varying delays are inevitable due to fluctuating network bandwidth, routing changes, and signal transmission uncertainty. Uyulan et al. tackled this challenge by proposing a robust passivity-based nonlinear controller equipped with a super-twisting sliding-mode observer and a disturbance observer to mitigate the effects of variable delays and load disturbances, ensuring stable coordination between master and slave manipulators [16]. Building on this robustness perspective, Chang et al. [17] introduced a robust adaptive sliding-mode control scheme that employs Lyapunov–Krasovskii functionals and LMI-based delay bounds to guarantee stability under bounded time-varying delays, validated through experimental results. Chen et al. [18] introduced a delay-adaptive synchronization controller that uses position-only communication, a velocity-feedback filter, and a radial basis function neural network. This approach addresses asymmetric time-varying delays and compensates for system uncertainties. Similarly, Lu et al. [13] proposed an adaptive control method for teleoperation systems with uncertain dynamics, focusing on parameter adaptation to maintain stability in the presence of delays and modeling errors. Chen et al. [19] advanced from adaptation to prediction by introducing a Smith predictive controller for space teleoperation. Their approach uses a delay estimation model to forecast communication latency and maintain transparency despite significant delay variations. Building on this, Ebrahimian et al. [20] integrated model-mediated environment force prediction with an artificial neural network that predicts the operator’s future motion, enabling real-time synchronization between master and slave systems even when delays are unpredictable. More recently, Dao et al. [21] addressed the same problem using a rise-based integral reinforcement learning (IRL) framework, where an actor–critic agent learns optimal control laws for systems with variable delays and disturbances, achieving both coordination and robustness. Meanwhile, Patiño et al. [22] employed neural network compensators in a skid-steer teleoperation system, where damping injection and learned interaction models jointly ensured stable motion tracking despite nonlinear time-delay effects. In a related approach, Xia et al. [23] reframed the problem from a learning standpoint by introducing a belief-state actor–critic (BSAC) algorithm that explicitly models action and observation delays within an agent-based space teleoperation framework, effectively mitigating both constant and random time lags. In a different teleoperation configuration involving one master and three slave manipulators, Khanzadeh et al. [24] introduced an adaptive synchronization controller designed to maintain coordination and stability under time-varying delays and formation changes. Incorporating joint or link flexibility into the teleoperation framework aligns system dynamics more closely with those of real-world manipulators. This increased realism introduces additional nonlinearities,

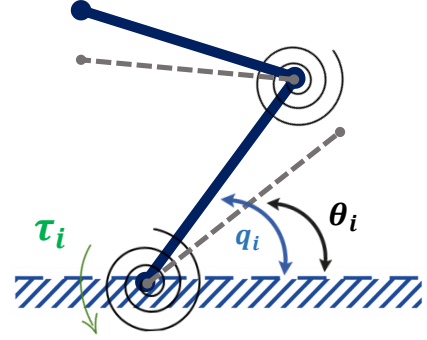


Fig. 1: Schematic of the 2-DOF flexible joint manipulator.

vibrations, and underactuation, which complicate the control problem, particularly in the presence of time delays. To address these challenges, several recent control strategies have been developed. Nuño et al. [25] developed adaptive and P+d controllers for teleoperation systems with joint flexibility and time delays, proving that both can ensure stable motion and asymptotic position tracking. By explicitly modeling joint elasticity, they showed that flexible manipulators more accurately represent real teleoperation dynamics while maintaining global stability through Lyapunov–Krasovskii analysis. Xu et al. [26] developed a TD3-based adaptive controller for a circulating cooling water system, showing that TD3 algorithms outperform PID, fuzzy PID, and DDPG methods in nonlinear environments. Their use of a reference trajectory model improved convergence and reduced oscillations, demonstrating the potential of deep reinforcement learning for systems with slow and uncertain dynamics. In parallel, Yin et al. [27] utilized a TD3 reinforcement learning controller for the vector control of permanent magnet synchronous motors (PMSM). By replacing the conventional PI controller within the current loop with a learned policy, they achieved faster response and reduced tracking error under variable operating conditions. Their results confirmed that DRL can effectively tune control gains online in complex continuous-action systems, a principle extendable to flexible manipulators. Barjini et al. [28] created a hybrid approach for flexible robotic arms that blends a physics-based controller with a learning-based motion planner. The controller keeps the system stable, while the planner uses soft actor–critic (SAC) methods to find movements that reduce unwanted vibrations. This combination led to better control and less shaking, as shown in lab tests on a flexible robot arm. Despite these advances, most existing studies focus on either compensating for time-varying delays or addressing joint flexibility. Few studies tackle both challenges simultaneously within a unified framework. Deep reinforcement learning shows promise in managing nonlinear and uncertain dynamics. However, its integration with bilateral teleoperation systems that include elastic joints and variable communication delays remains largely unexplored. Traditional adaptive and neural controllers still rely on manual gain tuning or precise model knowledge. This reliance limits their adaptability in unpredictable networked environments.

II. DYNAMIC MODEL OF THE BILATERAL TELEOPERATION SYSTEM

Firstly, gravity is eliminated in both the local and remote robots. After, we consider the flexible teleoperation system according to Fig.1, its non-linear dynamic behavior of the remote manipulator is given by [29], [30]:

$$M_r(q_r)\ddot{q}_r + C_r(q_r, \dot{q}_r)\dot{q}_r + S_r[q_r - \theta_r] = -\tau_e \quad (1)$$

$$J_r\ddot{\theta}_r + S_r[\theta_r - q_r] = \tau_r \quad (2)$$

The local is modeled as a manipulator with rigid joints. Its nonlinear behavior is governed by [10]:

$$M_l(q_l)\ddot{q}_l + C_l(q_l, \dot{q}_l)\dot{q}_l = \tau_h - \tau_l \quad (3)$$

Where $i \in \{l, r\}$ for local and remote manipulators, $q_i, \dot{q}_i, \ddot{q}_i \in \mathbb{R}^n$ denote the joint positions, velocities, and accelerations, and $\theta_r \in \mathbb{R}^n$ is the remote joint (motor) position, $M_i \in \mathbb{R}^{n \times n}$ are the inertia matrices. $C_i \in \mathbb{R}^{n \times n}$ represent the Coriolis and centrifugal effects, $J_r \in \mathbb{R}^{n \times n}$ is a constant diagonal matrix representing the remote actuator moments of inertia, $S_r \in \mathbb{R}^{n \times n}$ is a constant diagonal and positive definite matrix that contains the remote joint stiffness, $\tau_i \in \mathbb{R}^n$ are the control signals, and $\tau_h \in \mathbb{R}^n$, $\tau_e \in \mathbb{R}^n$ are the operator and environment torques. This model has the following properties [31].

- Property 1. The inertia matrices $M_i(q_i)$ are positive definite and bounded.
- Property 2. The matrix $\dot{M}_i(q_i) - 2C_i(q_i, \dot{q}_i)$ is skew-symmetric, $\dot{M}_i(q_i) = C_i(q_i, \dot{q}_i) + C_i^T(q_i, \dot{q}_i)$.

III. PROPOSED CONTROL STRATEGY

We introduce a hybrid control approach that combines a stable proportional plus damping injection ($P + d$) controller with a deep reinforcement learning (DRL) agent as an adaptive gain tuner. The $P + d$ controller ensures baseline stability, while the DRL agent continuously adjusts the proportional (K) and damping (B) gains in real time. This approach suppresses flexible-joint vibrations and maintains precise tracking despite unpredictable, time-varying communication delays, delivering greater robustness and adaptability compared to fixed-gain static controllers.

A. Proportional Plus Damping Injection ($P + d$) Controller

The $P + d$ controller serves as a core stabilizing element for bilateral teleoperation systems. This architecture has been shown in prior studies to ensure global stability for nonlinear teleoperators, establishing position tracking even in the presence of joint flexibility and variable time-varying delays. The specific control scheme implemented assumes a rigid local manipulator and a remote manipulator with flexible joints. Under this scheme, velocities and position errors remain bounded, and in the absence of external forces, exact position tracking is guaranteed. The formal stability guarantee for this control structure will be discussed in Section IV.

For the local (rigid) manipulator, feedback utilizes the link position $q_l(t)$ and link velocity $\dot{q}_l(t)$. For the remote (flexible) manipulator, only the joint (motor) position $\theta_r(t)$ and velocity

$\dot{\theta}_r(t)$ are used; the remote link position $q_r(t)$ is intentionally omitted due to the difficulty of measuring link-side states in flexible-joint systems.

Let $T_l(t)$ and $T_r(t)$ denote the time-varying communication delays on the forward (local-to-remote) and backward (remote-to-local) channels, respectively, with known upper bounds T_l^* and T_r^* . The delayed local link position received by the remote side is $q_l(t - T_l(t))$, and the delayed remote joint position received by the local side is $\theta_r(t - T_r(t))$.

The resulting $P + d$ control laws for the local torque τ_l and remote torque τ_r are:

$$\tau_l = K_l[q_l(t) - \theta_r(t - T_r(t))] + B_l\dot{q}_l(t) \quad (4)$$

$$\tau_r = -K_r[\theta_r(t) - q_l(t - T_l(t))] - B_r\dot{\theta}_r(t) \quad (5)$$

In these control laws, the proportional gains K_l and K_r correct position errors by applying force proportional to how far the system is from its target. The damping gains B_l and B_r add resistance to motion, smoothing movement and reducing oscillations. All gain terms are assumed to be positive constants.

B. DRL Agents for Adaptive Gain Tuning

Conventional controllers often depends on fixed gains from linearized models or manual adjustment, causing these methods to struggle with system nonlinearities, uncertain parameters, or dynamics that change over time. DRL directly addresses these limitations by adapting the controller gains online without any prior model. As a model-free approach, DRL eliminates the need for system identification, offline tuning, or expert knowledge. Fundamentally, DRL operates as a decision-making agent that learns an optimal policy (a mapping from states of the system to actions) by maximizing a cumulative reward signal through trial and error. It is naturally suited for continuous control tasks like gain tuning, as it can directly output real-valued control actions (the gains) and learn stable control policies purely through environmental interaction. By optimizing rewards based on closed-loop performance (e.g., minimizing tracking and damping error), the DRL agent continuously refines the control parameters in real time. This ensures robust control performance even when facing disturbances, setpoint changes, or shifts in operating conditions, making it particularly effective for complex or uncertain systems where traditional methods are inadequate [27] [32].

C. DRL Problem Formulation

To implement a DRL agent for adaptive gain tuning, the control problem is formalized within the standard Markov Decision Process (MDP) framework. This involves defining the core components: the state representation, the action space, and the reward function. To simplify the training process, the DRL agent is implemented exclusively on the remote controller, while local control is omitted in this study and left for future work.

- **State (s_t):** The state vector offers the agent a complete observation of the system's dynamic condition at each

time step t . It is formulated according to the teleoperator dynamics as follows:

$$\mathbf{s}_t = [\theta_r(t), \dot{\theta}_r(t), q_l(t - T_l(t)), e_{\text{tracking}}(t), \int_0^t e_{\text{tracking}}(\tau) d\tau, e_{\text{damping}}(t), \int_0^t e_{\text{damping}}(\tau) d\tau]^T, \quad (6)$$

where θ_r and $\dot{\theta}_r$ denote the remote joint (motor) position and velocity, respectively; $q_l(t - T_l(t))$ represents the delayed local link position; $e_{\text{tracking}} = \theta_r - q_l(t - T_l(t))$ is the tracking error between the remote and delayed local positions; $\int e_{\text{tracking}} d\tau$ is its integral; $e_{\text{damping}} = \theta_r - q_l$ represents the flexibility (damping) error between the remote link position (q_r) and the remote joint (motor) position; and $\int e_{\text{damping}} d\tau$ denotes its integral.

- **Action ($\mathbf{a}_t$):** The action $\mathbf{a}_t$ represents the output of the policy network, which determines the adaptive gain tuning for the remote controller. Specifically, the action defines the proportional gain (K) and damping injection gain vector (B) for each degree of freedom (DOF). It is expressed as

$$\mathbf{a}_t = [K_r^i, B_r^i], \quad i = \{1, 2, \dots, n\}, \quad (7)$$

where K_r^i and B_r^i denote the proportional and damping gains of the i_{th} joint of the remote manipulator, and n is the total number of joints (motors). The gains of the local manipulator are kept constant and are not adaptively tuned.

- **Reward (r_t):** The reward function quantifies the control performance and guides the agent toward the desired behavior. The reward at time step t is defined as

$$r_t = -(\alpha \|e_{\text{tracking}}\|^2 + \beta \|e_{\text{damping}}\|^2 + \gamma \|\mathbf{a}_t\|^2), \quad (8)$$

where α , β , and γ are positive weighting coefficients that balance the contribution of each term. This formulation penalizes large tracking and damping errors through the first two terms, while $\|\mathbf{a}_t\|^2$ penalizes excessive changes in the controller gains to promote smooth and stable tuning.

By interacting with the bilateral teleoperation environment as illustrated in Fig. 2(a), the DRL agent observes the system state $\mathbf{s}_t$, takes an action $\mathbf{a}_t = [K_r^i, B_r^i]$ to adaptively adjust the proportional and damping gains of the remote controller, and receives a reward r_t based on the resulting tracking and damping performance.

D. Twin Delayed Deep Deterministic Policy Gradient (TD3)

The Twin Delayed Deep Deterministic Policy Gradient (TD3) [33] algorithm is a model-free and off-policy deep reinforcement learning method that extends the Deep Deterministic Policy Gradient (DDPG) framework for continuous control [34]. Its architecture comprises six neural networks: an online and target Actor, along with two distinct pairs of

online and target Critics. TD3 improves upon DDPG by addressing its systematic issue of Q-value overestimation through three key mechanisms. First, it employs twin Critic networks, utilizing the minimum of their Q-value estimates to reduce overestimation bias. Second, it delays Actor updates, ensuring that policy improvements are guided by a more stable value function. Third, it applies target policy smoothing via additive, clipped Gaussian noise to target actions during Critic updates, which regularizes the policy and mitigates the exploitation of value estimation errors. The structure of the TD3 algorithm can be found in Fig. 2(b).

IV. STABILITY ANALYSIS

We consider a rigid local manipulator and a remote manipulator with joint flexibility (only on the remote side). Let the link and joint states be q_l , q_r , and θ_r . The communication delays on the forward and backward channels are time-varying and bounded:

$$0 \leq T_i(t) \leq T_i^*, \quad i \in \{l, r\}. \quad (9)$$

Human and environment operators are modeled as passive mappings from velocity to torque:

$$\begin{aligned} E_h &= -\int_0^t \dot{q}_l^\top(\sigma) \tau_h(\sigma) d\sigma + \kappa_l \geq 0, \\ E_e &= \int_0^t \dot{q}_r^\top(\sigma) \tau_e(\sigma) d\sigma + \kappa_r \geq 0. \end{aligned} \quad (10)$$

The delayed coordination errors (for the variable-delay case) are defined as

$$\begin{aligned} e_l &= q_l - \theta_r(t - T_r(t)), \\ e_r &= \theta_r - q_l(t - T_l(t)). \end{aligned} \quad (11)$$

The local and remote torques follow a P+d structure with time-varying gains provided by the TD3 agent:

$$\begin{aligned} \tau_l &= K_l(t) e_l + B_l(t) \dot{q}_l, \\ \tau_r &= -K_r(t) e_r - B_r(t) \dot{\theta}_r. \end{aligned} \quad (12)$$

The TD3 outputs are constrained to an admissible set $\mathcal{S}$ to preserve stability:

$$K_i^{\min} \leq K_i(t) \leq K_i^{\max}, \quad B_i^{\min} \leq B_i(t) \leq B_i^{\max}. \quad (13)$$

The delay-robustness inequality holds for all t :

$$4 B_l(t) B_r(t) \geq (T_l^* + T_r^*)^2 K_l(t) K_r(t) + \varepsilon, \varepsilon > 0, \quad (14)$$

where ε provides a safety margin.

A. Lyapunov–Krasovskii Functional

We adopt the same Lyapunov–Krasovskii functional as in the paper by Nuno [25], using constant positive weights independent of time-varying gains:

$$V = V_l + \eta V_r + E_h + \eta E_e + \frac{\eta K_l}{2} |q_l - \theta_r|, \quad (15)$$

$$\begin{aligned} V_r(q_r, \dot{q}_r, \theta_r, \dot{\theta}_r) &= \frac{1}{2} (\dot{q}_r^\top M_r(q_r) \dot{q}_r + \dot{\theta}_r^\top J_r \dot{\theta}_r) \\ &\quad + \frac{1}{2} (q_r - \theta_r)^\top S_r (q_r - \theta_r). \end{aligned} \quad (16)$$

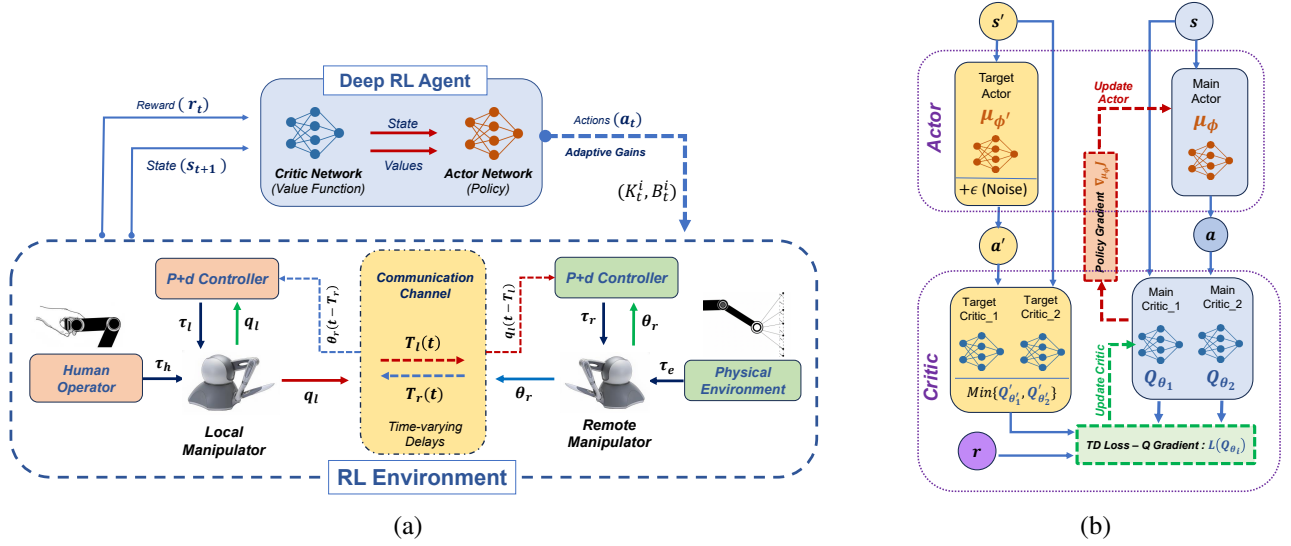


Fig. 2: Overview of the proposed control framework: (a) bilateral teleoperation system with joint flexibility and time-varying delays, and (b) TD3 network architecture used by the DRL agent for adaptive gain tuning.

$$V_l(q_l, \dot{q}_l) = \frac{1}{2} \dot{q}_l^T M_l(q_l) \dot{q}_l, \quad (17)$$

where V_l , V_r represent the kinetic and potential energies of the local rigid and remote flexible subsystems, $\eta > 0$ is a fixed scalar, and K_l is a known lower bound.

For variable delays, the following integral inequality, stated as Lemma 1 in [25], is repeatedly used to prove stability:

$$-\int_0^t x^T(\sigma) \int_{-T(\sigma)}^0 y(\sigma + \theta) d\theta d\sigma \leq \frac{\alpha}{2} \|x\|_2^2 + \frac{(T^*)^2}{2\alpha} \|y\|_2^2. \quad (18)$$

Differentiating V along (12) and applying (18) yields

$$\dot{V} \leq -\lambda_l \|\dot{q}_l\|^2 - \lambda_r \|\dot{\theta}_r\|^2, \quad (19)$$

where $\lambda_l, \lambda_r > 0$ exist if and only if (14) holds.

Remark: Since V uses fixed positive weights, no $\dot{K}$ or $\dot{B}$ terms appear in $\dot{V}$. Hence, TD3 may vary the gains arbitrarily fast as long as they remain inside $\mathcal{S}$.

V. NUMERICAL SIMULATION & RESULTS

The numerical simulations were conducted to evaluate the effectiveness of the proposed adaptive DRL control strategy under challenging teleoperation conditions involving joint flexibility, environmental uncertainties, and time-varying communication delays. All simulations were implemented in MATLAB/Simulink over a duration of $T_f = 40$ s with a sampling time of $T_s = 0.01$ s.

A. Bilateral Teleoperation Setup

The system comprises a rigid 2-DOF master (local) manipulator and a flexible-joint 2-DOF slave (remote) manipulator, with dynamics detailed in Section II. Joint flexibility on the remote side is modeled via link positions q_r and motor positions θ_r , coupled by a diagonal stiffness matrix $S_r = 100I$ N-m.

Physical parameters include link lengths $l_1 = l_2 = 0.38$ m, link masses $m_{1l} = m_{1r} = 0.5$ kg and $m_{2l} = m_{2r} = 0.35$ kg, and remote motor inertia $J_r = 0.3I$ kg-m². The local controller uses fixed gains $K_l = 25$ N-m and $B_l = 10$ N-m-s.

Time-varying communication delays $T_i(t)$ are bounded by $T_l^* = T_r^* = 1$ s (see Fig. 3). The human operator is modeled as $\tau_h = K_s(q_h - q_l) - K_d \dot{q}_l$, with $K_s = 10$ N/m, $K_d = 2$ N-s/m, and q_h a smooth sinusoidal reference. The environment torque is set to $\tau_e = 0$, representing free-space operation.

B. DRL Training Details

The TD3 agent was trained within a simulated bilateral teleoperation environment to adaptively tune the proportional and damping gains of the remote controller. The actor and critic networks were implemented as fully connected multilayer perceptrons with ReLU activations in the hidden layers. The actor's final layer used a Softplus activation function to ensure all adaptive gains remained positive and smooth. Each critic network received both state and action inputs, which were processed through parallel feature-extraction layers before being merged to estimate the Q-value. Two critic networks were employed to reduce value overestimation and stabilize learning.

The main hyperparameters were set as follows: the discount factor $\gamma = 0.99$, minibatch size of 512, and learning rates of 1×10^{-4} for both the actor and critics, using the Adam optimizer with gradient clipping of 1. Gaussian exploration noise was applied with an initial standard deviation of 4, decaying exponentially to a minimum of 0.05 at a rate of 1×10^{-7} . The target networks were softly updated with coefficient $\tau = 0.005$.

C. Control Performance Evaluation

The proposed DRL-based adaptive gain tuning controller demonstrates robust performance in stabilizing the flexible-

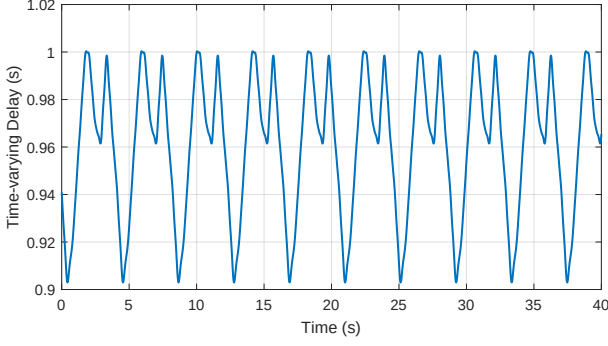


Fig. 3: Communication channel delay profile.

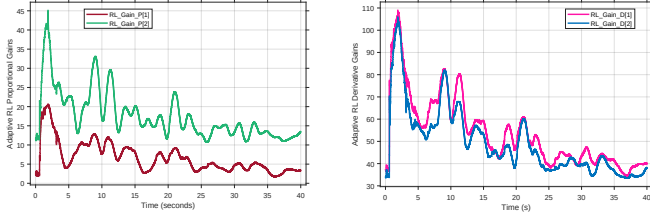


Fig. 4: DRL-generated adaptive gains for $i = 1, 2$: (left) proportional gains K_t^i , and (right) derivative gains B_t^i .

joint teleoperation system while achieving high tracking accuracy, even in the presence of joint flexibility and time-varying communication delays.

As shown in Fig. 5, the time-domain response of the system reveals that the DRL controller effectively suppresses the oscillatory behavior inherent in flexible-joint systems with delays. Both the remote joint positions (θ_r) and the delayed local link positions (q_l) rapidly converge to a synchronized steady state, indicating strong stabilization capabilities.

Fig. 6 presents the position tracking error, defined as $\theta_r - q_l(t - T_l)$, where T_l denotes the forward communication delay. For both joints, the tracking error converges swiftly to a neighborhood of zero, underscoring the controller's ability to maintain tight coordination between the master and slave sides despite time-varying delays. The negligible steady-state error further attests to the efficacy of the DRL framework in compensating for such uncertainties.

Moreover, Fig. 7 illustrates the remote link–joint damping error ($\theta_r - q_r$), which reflects the level of vibration induced by joint flexibility. The rapid attenuation of this error for both joints confirms that the adaptive gain tuning mechanism successfully emulates virtual damping, thereby mitigating the destabilizing influence of joint elasticity.

In summary, the experimental results validate that the proposed DRL-based adaptive controller achieves fast stabilization, accurate position tracking, and effective vibration suppression in a flexible-joint teleoperation system, successfully addressing the dual challenges posed by mechanical compliance and time-varying network delays.

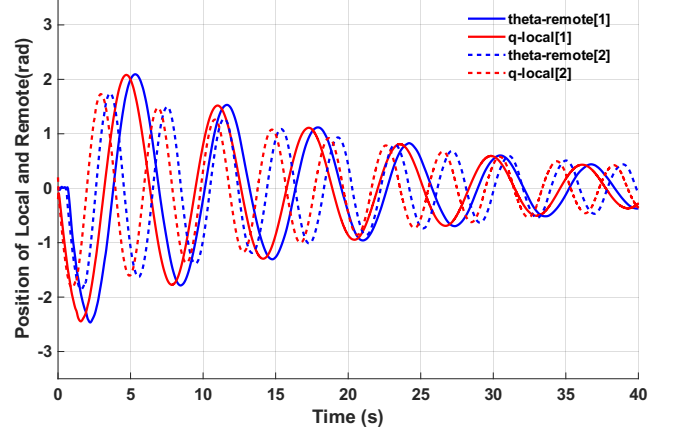


Fig. 5: Time response of remote joint positions $\theta_r^i(t)$ and local link positions $q_l^i(t - T_l)$ for $i = 1, 2$.

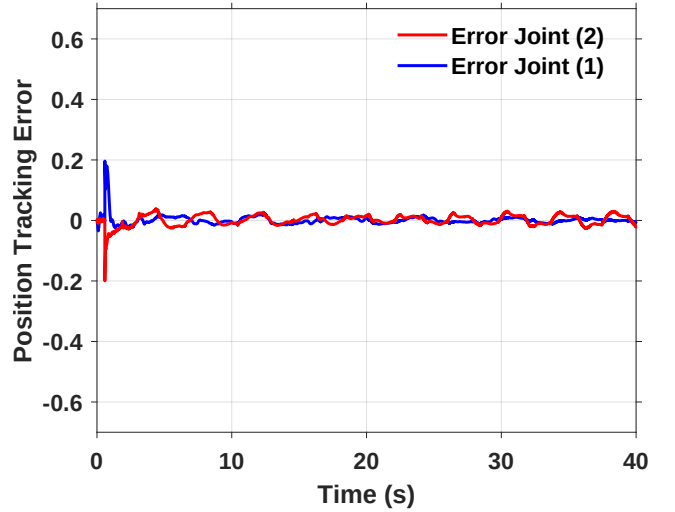


Fig. 6: Tracking error between the delayed local link position $q_l(t - T_l)$ and the remote joint angle $\theta_r(t)$.

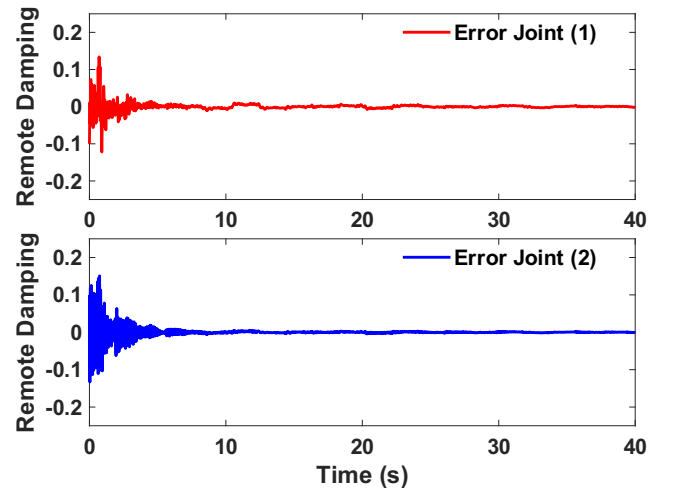


Fig. 7: Remote link–joint damping error ($\theta_r - q_r$)

VI. CONCLUSION AND FUTURE WORK

This work presented a hybrid control framework that combines a stable P+d controller with a TD3-based DRL agent to address joint flexibility and time-varying delays in bilateral teleoperation. The DRL agent adaptively tunes the remote-side gains in real time, while Lyapunov–Krasovskii analysis guarantees stability under bounded delays. Simulations on a 2-DOF flexible manipulator with delays up to 1s confirmed that the approach achieves fast stabilization, accurate position tracking, and effective vibration suppression, successfully handling the dual challenges of joint flexibility and uncertain network delays. Future work will extend adaptive gain tuning to the master side, incorporate impedance adaptation for interactions with stiff environments, explore a multi-agent framework [35] to coordinate independent reinforcement learning on both master and slave sides, validate the approach on a physical teleoperation platform under real-world conditions, and integrate sim-to-real techniques [36].

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